Advanced Statistical Methods II

Non-linear Regression: Two Problems with Solutions

1 Problem 1

Consider the following model:

$$y_t = \frac{1}{1+\theta t} + \epsilon_t.$$

You have a sample of size n, say $1 < t_1 < \cdots < t_n < 10$. Here ϵ_t's are independent and identically distributed normal random variables with mean zero and finite variance σ^2 and $\theta > -1/10$.

(a) Show that the maximum likelihood estimator of θ can be obtained by solving a one-dimensional optimization problem.

(b) Provide the Newton–Raphson algorithm to find the maximum likelihood estimator of θ from an initial guess value say $\theta^{(0)}$.

(c) Find an approximate 95% confidence interval of θ.

(d) Using (c) or otherwise, test the following hypothesis $H_0 : \theta = \theta_0$ vs. $H_1 : \theta \neq \theta_0$.

(e) For a given n (quite large), find t_i's so that $1 < t_1 < \cdots < t_n < 10$, and it produces approximately the most *efficient* (minimum variance) maximum likelihood estimator of θ.

Notations

Write the observations as $y_i = y_{t_i}$, $i = 1, \ldots, n$, and put

$$f_i(\theta) = \frac{1}{1+\theta t_i}, \qquad u_i(\theta) = 1+\theta t_i, \qquad r_i(\theta) = y_i - f_i(\theta) = y_i - \frac{1}{u_i}, \qquad S(\theta) = \sum_{i=1}^{n} r_i(\theta)^2.$$

We will need the first two derivatives of the mean function. Since $f_i = u_i^{-1}$ and $du_i/d\theta = t_i$,

$$f_i'(\theta) = \frac{d}{d\theta}(1+\theta t_i)^{-1} = -t_i(1+\theta t_i)^{-2} = -\frac{t_i}{u_i^2},$$
$$f_i''(\theta) = \frac{d}{d\theta}\{-t_i(1+\theta t_i)^{-2}\} = 2t_i^2(1+\theta t_i)^{-3} = \frac{2t_i^2}{u_i^3}.$$

1.1 (a) The MLE solves a one-dimensional problem

Since $y_i \sim N(f_i(\theta), \sigma^2)$ independently, the likelihood is

$$L(\theta, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big\{-\frac{(y_i - f_i(\theta))^2}{2\sigma^2}\Big\},$$

and the log-likelihood is

$$\ell(\theta, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\sigma^2 - \frac{S(\theta)}{2\sigma^2}.$$

There are two unknowns, θ and σ^2. We maximize over σ^2 first, for fixed θ.

Step 1: maximize over σ^2.

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{S(\theta)}{2\sigma^4} = 0 \quad \Longleftrightarrow \quad \sigma^2 = \widehat{\sigma}^2(\theta) := \frac{S(\theta)}{n}.$$

This is a maximum, because

$$\frac{\partial^2 \ell}{\partial (\sigma^2)^2}\Big|_{\sigma^2=\widehat{\sigma}^2(\theta)} = \frac{n}{2\widehat{\sigma}^4} - \frac{S(\theta)}{\widehat{\sigma}^6} = \frac{n}{2\widehat{\sigma}^4} - \frac{n}{\widehat{\sigma}^4} = -\frac{n}{2\widehat{\sigma}^4} < 0.$$

(With probability one $S(\theta) > 0$, since the y_i have a continuous distribution and cannot all lie exactly on the curve.)

Step 2: the profile log-likelihood. Substituting $\widehat{\sigma}^2(\theta)$,

$$\begin{aligned}\ell_p(\theta) := \ell\big(\theta, \widehat{\sigma}^2(\theta)\big) &= -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\frac{S(\theta)}{n} - \frac{S(\theta)}{2S(\theta)/n} \\ &= -\frac{n}{2}\log S(\theta) + \underbrace{\Big\{-\frac{n}{2}\log(2\pi) + \frac{n}{2}\log n - \frac{n}{2}\Big\}}_{\text{free of } \theta}.\end{aligned}$$

Step 3: reduce to one dimension. Since

$$\sup_{\theta,\sigma^2} \ell(\theta, \sigma^2) = \sup_{\theta}\sup_{\sigma^2} \ell(\theta, \sigma^2) = \sup_{\theta} \ell_p(\theta),$$

and $-\frac{n}{2}\log S$ is a strictly decreasing function of S,

$$\widehat{\theta}_{\mathrm{ML}} = \arg\max_{\theta} \ell_p(\theta) = \arg\min_{\theta} S(\theta) = \arg\min_{\theta} \sum_{i=1}^{n}\Big\{y_i - \frac{1}{1+\theta t_i}\Big\}^2, \qquad \widehat{\sigma}^2_{\mathrm{ML}} = \frac{S(\widehat{\theta})}{n}.$$

So the MLE of θ is the non-linear least-squares estimator, and it is obtained by minimizing a function of the *single* variable θ. Setting $S'(\theta) = 0$ gives the likelihood (normal) equation

$$S'(\theta) = -2\sum_{i=1}^{n} r_i f_i' = 2\sum_{i=1}^{n} \frac{t_i}{u_i^2}\Big(y_i - \frac{1}{u_i}\Big) = 0. \tag{1}$$

1.2 (b) The Newton–Raphson algorithm

Derivatives of S. Using $\partial r_i/\partial\theta = -f_i'$,

$$S'(\theta) = 2\sum_i r_i \frac{\partial r_i}{\partial \theta} = -2\sum_i r_i f_i', \qquad S''(\theta) = -2\sum_i \Big(\frac{\partial r_i}{\partial\theta} f_i' + r_i f_i''\Big) = 2\sum_i \big(f_i'^2 - r_i f_i''\big).$$

Now substitute $f_i' = -t_i/u_i^2$ and $f_i'' = 2t_i^2/u_i^3$:

$$S'(\theta) = -2\sum_i \Big(y_i - \frac{1}{u_i}\Big)\Big(-\frac{t_i}{u_i^2}\Big) = 2\sum_i \frac{t_i}{u_i^2}\Big(y_i - \frac{1}{u_i}\Big),$$

$$S''(\theta) = 2\sum_i \Big\{\frac{t_i^2}{u_i^4} - \Big(y_i - \frac{1}{u_i}\Big)\frac{2t_i^2}{u_i^3}\Big\} = 2\sum_i \Big\{\frac{t_i^2}{u_i^4} - \frac{2t_i^2 y_i}{u_i^3} + \frac{2t_i^2}{u_i^4}\Big\} = 2\sum_i \frac{t_i^2\,(3 - 2y_i u_i)}{u_i^4}.$$

Newton–Raphson for the likelihood equation. For fixed σ^2, $\ell(\theta) = \text{const} - S(\theta)/(2\sigma^2)$, so

$$\frac{\partial \ell}{\partial \theta} = -\frac{S'(\theta)}{2\sigma^2}, \qquad \frac{\partial^2 \ell}{\partial \theta^2} = -\frac{S''(\theta)}{2\sigma^2}.$$

Newton–Raphson solves $\partial\ell/\partial\theta = 0$ by linearizing the score around the current iterate: $\ell'(\theta) \approx \ell'(\theta^{(k)}) + \ell''(\theta^{(k)})(\theta - \theta^{(k)}) = 0$. Hence

$$\theta^{(k+1)} = \theta^{(k)} - \frac{\ell'(\theta^{(k)})}{\ell''(\theta^{(k)})} = \theta^{(k)} - \frac{-S'(\theta^{(k)})/(2\sigma^2)}{-S''(\theta^{(k)})/(2\sigma^2)} = \theta^{(k)} - \frac{S'(\theta^{(k)})}{S''(\theta^{(k)})}.$$

The unknown σ^2 cancels, so the iteration does not need it. Writing $u_i^{(k)} = 1 + \theta^{(k)} t_i$ and cancelling the factor 2,

$$\boxed{\theta^{(k+1)} = \theta^{(k)} - \frac{\displaystyle\sum_{i=1}^{n} \frac{t_i}{(u_i^{(k)})^2}\Big(y_i - \frac{1}{u_i^{(k)}}\Big)}{\displaystyle\sum_{i=1}^{n} \frac{t_i^2\,\big(3 - 2y_i u_i^{(k)}\big)}{(u_i^{(k)})^4}}} \tag{2}$$

The algorithm.

Step 1. *Initial value.* Choose $\theta^{(0)}$ and a tolerance τ (say 10^{-8}). A good automatic choice comes from linearizing the model: if $y_i \approx 1/(1 + \theta t_i)$ then $1/y_i - 1 \approx \theta t_i$, a line through the origin, so

$$\theta^{(0)} = \frac{\sum_{i:\, y_i > 0} t_i\,(1/y_i - 1)}{\sum_{i:\, y_i > 0} t_i^2}.$$

Step 2. *Compute* $u_i^{(k)}$, $S'(\theta^{(k)})$ and $S''(\theta^{(k)})$ from the formulas above.

Step 3. *Newton step.* If $S''(\theta^{(k)}) > 0$, set $\delta = -S'(\theta^{(k)})/S''(\theta^{(k)})$, as in (2). If $S''(\theta^{(k)}) \le 0$ the Newton step may point uphill; replace S'' by its expectation $2\sum_i f_i'^2 = 2\sum_i t_i^2/u_i^4 > 0$ (Fisher scoring, which is the Gauss–Newton step):

$$\delta = -\frac{\sum_i t_i r_i/u_i^2}{\sum_i t_i^2/u_i^4}.$$

Step 4. *Safeguard.* While $S(\theta^{(k)} + \delta) > S(\theta^{(k)})$ or $1 + (\theta^{(k)} + \delta)t_i \le 0$ for some i, halve δ.

Step 5. *Update and stop.* Set $\theta^{(k+1)} = \theta^{(k)} + \delta$. Stop if $|\theta^{(k+1)} - \theta^{(k)}| < \tau(|\theta^{(k)}| + \tau)$; otherwise go back to Step 2.

Step 6. *Output* $\widehat{\theta} = \theta^{(k+1)}$ and $\widehat{\sigma}^2 = S(\widehat{\theta})/n$.

Near the solution $S''(\widehat{\theta}) > 0$, no safeguard is triggered, and the iteration converges quadratically.

Why the safeguard is needed. In $S'' = 2\sum_i (f_i'^2 - r_i f_i'')$ the second term can dominate when the residuals are large, that is, far from the solution. Then $S'' < 0$ and the pure Newton step moves towards a maximum of S rather than a minimum.

1.3 (c) An approximate 95% confidence interval

The Fisher information. For one observation, $\ell_i = -\frac{1}{2}\log(2\pi\sigma^2) - r_i^2/(2\sigma^2)$, so

$$-\frac{\partial^2 \ell_i}{\partial \theta^2} = \frac{f_i'^2 - r_i f_i''}{\sigma^2}, \qquad \mathbb{E}\Big(-\frac{\partial^2 \ell_i}{\partial \theta^2}\Big) = \frac{f_i'^2}{\sigma^2},$$

since $\mathbb{E}(r_i) = 0$ at the true parameter. Moreover $\partial^2 \ell / \partial\theta\, \partial\sigma^2 = -\sum_i r_i f_i'/\sigma^4$ has expectation zero, so θ and σ^2 are orthogonal and the information for θ is

$$I_n(\theta) = \frac{1}{\sigma^2}\sum_{i=1}^{n} f_i'(\theta)^2 = \frac{1}{\sigma^2}\sum_{i=1}^{n} \frac{t_i^2}{(1+\theta t_i)^4}.$$

Asymptotic normality. The design points lie in $(1, 10)$ and, for $\theta^* > -1/10$, the denominators $1 + \theta^* t_i$ are bounded away from zero. Assume that $\frac{1}{n}\sum_i t_i^2/(1+\theta^* t_i)^4 \to c(\theta^*) > 0$; this holds, for example, when the t_i spread over $(1, 10)$ according to a fixed design density. Then the standard theory of maximum-likelihood (equivalently, non-linear least-squares) estimation gives

$$\sqrt{n}\,(\widehat{\theta} - \theta^*) \xrightarrow{d} N\Big(0, \frac{\sigma^2}{c(\theta^*)}\Big), \qquad \text{that is,} \qquad \widehat{\theta} \;\dot{\sim}\; N\Big(\theta^*,\ \frac{\sigma^2}{\sum_i t_i^2/(1+\theta^* t_i)^4}\Big).$$

Plug-in standard error. Replace θ^* by $\widehat{\theta}$ and σ^2 by an estimate:

$$\mathrm{se}(\widehat{\theta}) = \frac{\widehat{\sigma}}{\sqrt{\sum_{i=1}^{n} t_i^2/(1+\widehat{\theta} t_i)^4}}.$$

Since $\widehat{\theta}$ and $\widehat{\sigma}$ are consistent, Slutsky's theorem gives $Z = (\widehat{\theta} - \theta^*)/\mathrm{se}(\widehat{\theta}) \xrightarrow{d} N(0,1)$, so $P(-1.96 \le Z \le 1.96) \to 0.95$. Solving the inequalities for θ^* gives the approximate 95% confidence interval

$$\boxed{\widehat{\theta} \pm 1.96\ \mathrm{se}(\widehat{\theta}) = \left[\widehat{\theta} - 1.96\,\frac{\widehat{\sigma}}{\sqrt{\sum_i t_i^2/(1+\widehat{\theta} t_i)^4}},\ \widehat{\theta} + 1.96\,\frac{\widehat{\sigma}}{\sqrt{\sum_i t_i^2/(1+\widehat{\theta} t_i)^4}}\right].}$$

In moderate samples it is usual to take $\widehat{\sigma}^2 = s^2 = S(\widehat{\theta})/(n-1)$ and replace 1.96 by $t_{n-1,0.975}$. One may also use the *observed* information $S''(\widehat{\theta})/(2\widehat{\sigma}^2)$ in place of $I_n(\widehat{\theta})$; the two agree to first order.

1.4 (d) Testing $H_0 : \theta = \theta_0$ against $H_1 : \theta \neq \theta_0$

Using the interval of (c). Reject H_0 at level 5% if θ_0 is *not* in the interval of (c). Since

$$\theta_0 \in \left[\widehat{\theta} - 1.96\ \mathrm{se}(\widehat{\theta}),\ \widehat{\theta} + 1.96\ \mathrm{se}(\widehat{\theta})\right] \iff |\widehat{\theta} - \theta_0| \leq 1.96\ \mathrm{se}(\widehat{\theta}),$$

this is the *Wald test*: with

$$Z = \frac{\widehat{\theta} - \theta_0}{\mathrm{se}(\widehat{\theta})} = \frac{(\widehat{\theta} - \theta_0)\sqrt{\sum_i t_i^2/(1+\widehat{\theta} t_i)^4}}{\widehat{\sigma}},$$

reject H_0 if $|Z| > 1.96$. The approximate p-value is $2\{1-\Phi(|Z|)\}$. Under H_0, $Z \xrightarrow{d} N(0,1)$ by (c), so $P_{\theta_0}(|Z| > 1.96) \to 0.05$: the test has asymptotic size 5%. Under any fixed $\theta^* \neq \theta_0$, $|Z| \to \infty$ in probability, so the test is consistent.

1.5 (e) The most efficient design

The objective. By (c), for large n

$$\mathrm{Var}(\widehat{\theta}) \approx \frac{\sigma^2}{\sum_{i=1}^n g(t_i)}, \qquad g(t) = g(t;\theta) = \frac{t^2}{(1+\theta t)^4}.$$

Minimizing the variance means maximizing $\sum_i g(t_i)$ over $1 < t_1 < \cdots < t_n < 10$. Each term satisfies $g(t_i) \leq \sup_{1<t<10} g(t)$, so

$$\sum_{i=1}^n g(t_i) \leq n \sup_{1<t<10} g(t),$$

and the bound is approached by putting *all* design points at (or as close as possible to) the maximizer t^* of g. So the problem reduces to maximizing the function g of one variable.

Maximizing g. Take logarithms (with $t > 0$ and $1 + \theta t > 0$):

$$\log g(t) = 2\log t - 4\log(1+\theta t),$$

$$\frac{d}{dt}\log g(t) = \frac{2}{t} - \frac{4\theta}{1+\theta t} = \frac{2(1+\theta t) - 4\theta t}{t(1+\theta t)} = \frac{2(1-\theta t)}{t(1+\theta t)}.$$

The denominator is positive, so g is increasing where $\theta t < 1$ and decreasing where $\theta t > 1$. Three cases arise.

- $-0.1 < \theta \leq 0.1$: $\theta t < 1$ for all $t \in (1, 10)$, so g is increasing and $t^* = 10$. Put the points just below 10.
- $0.1 \leq \theta \leq 1$: g increases up to $t = 1/\theta \in [1, 10]$ and decreases after it, so

$$t^* = \frac{1}{\theta}, \qquad g(t^*) = \frac{1/\theta^2}{(1+1)^4} = \frac{1}{16\,\theta^2}, \qquad \mathrm{Var}(\widehat{\theta}) \approx \frac{\sigma^2}{n\,g(t^*)} = \frac{16\,\theta^2\sigma^2}{n}.$$

- $\theta \geq 1$: g is decreasing on $(1, 10)$, so $t^* = 1$. Put the points just above 1.

The design. Because the t_i must be distinct, take them in a small interval around t^* inside $(1, 10)$, for example

$$t_i = t^* + \Big(i - \frac{n+1}{2}\Big)\delta, \qquad i = 1, \dots, n,$$

with $\delta > 0$ small (and shifted inward if t^* is an end point). Since g is continuous, $\sum_i g(t_i) \to n\, g(t^*)$ as $\delta \to 0$, so this design is approximately the most efficient.

2 Problem 2

Consider the following regression model:

$$y_t = \alpha + \beta t + A\cos(\omega t^2) + B\sin(\omega t^2) + \epsilon_t.$$

You have a sample of size n, say $y_1, \dots, y_n$. Here ϵ_t's are i.i.d (mean $= 0$, variance $= \sigma^2$). $-\infty < \alpha, \beta, A, B < \infty$ and $\omega \in [0, \pi]$.

(a) Show that the least squares estimators of the unknown parameters can be obtained by solving a one-dimensional optimization problem.

(b) If you have the solution of the one-dimensional problem of (a), find the least squares estimators of the other parameters in terms of that solution.

(c) Suppose you know ω, for what values of ω all the other parameters can be estimated.

(d) Assuming ω is known, find a 95% confidence set of the other parameters.

Notation

The observations are taken at $t = 1, 2, \dots, n$. For a given ω write

$$c_t = c_t(\omega) = \cos(\omega t^2), \qquad s_t = s_t(\omega) = \sin(\omega t^2), \qquad \boldsymbol{\gamma} = (\alpha, \beta, A, B)^{\mathsf{T}},$$

$$\mathbf{X}(\omega) = \begin{pmatrix} 1 & 1 & c_1 & s_1 \\ 1 & 2 & c_2 & s_2 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & n & c_n & s_n \end{pmatrix} \quad (n \times 4), \qquad \mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}.$$

Then the model is $\mathbf{y} = \mathbf{X}(\omega)\boldsymbol{\gamma} + \boldsymbol{\varepsilon}$, and the least-squares criterion is

$$Q(\boldsymbol{\gamma}, \omega) = \sum_{t=1}^{n} \big(y_t - \alpha - \beta t - Ac_t - Bs_t\big)^2 = \|\mathbf{y} - \mathbf{X}(\omega)\boldsymbol{\gamma}\|^2.$$

The key observation is that the model is *linear* in $\boldsymbol{\gamma}$ once ω is fixed.

Remark 1 (Why $\omega \in [0, \pi]$). Since t^2 is an integer, replacing ω by $\omega + 2\pi$ leaves c_t and s_t unchanged, and replacing ω by $2\pi - \omega$ gives $c_t(2\pi - \omega) = c_t(\omega)$ and $s_t(2\pi - \omega) = -s_t(\omega)$, which is absorbed by changing B to $-B$. So without a restriction ω would not be identifiable; $[0, \pi]$ is a fundamental range.

2.1 (a) Reduction to a one-dimensional problem

Step 1: fix ω and minimize over $\boldsymbol{\gamma}$. For fixed ω, write $\mathbf{X} = \mathbf{X}(\omega)$ and suppose it has full column rank 4 (part (c) says when). Expanding,

$$Q(\boldsymbol{\gamma}, \omega) = \mathbf{y}^\mathsf{T}\mathbf{y} - 2\boldsymbol{\gamma}^\mathsf{T}\mathbf{X}^\mathsf{T}\mathbf{y} + \boldsymbol{\gamma}^\mathsf{T}\mathbf{X}^\mathsf{T}\mathbf{X}\boldsymbol{\gamma}, \quad \frac{\partial Q}{\partial \boldsymbol{\gamma}} = -2\mathbf{X}^\mathsf{T}\mathbf{y} + 2\mathbf{X}^\mathsf{T}\mathbf{X}\boldsymbol{\gamma}, \quad \frac{\partial^2 Q}{\partial \boldsymbol{\gamma}\, \partial \boldsymbol{\gamma}^\mathsf{T}} = 2\mathbf{X}^\mathsf{T}\mathbf{X}.$$

The Hessian $2\mathbf{X}^\mathsf{T}\mathbf{X}$ is positive definite, so $Q(\cdot, \omega)$ is strictly convex in $\boldsymbol{\gamma}$, and its unique minimizer solves the normal equations $\mathbf{X}^\mathsf{T}\mathbf{X}\boldsymbol{\gamma} = \mathbf{X}^\mathsf{T}\mathbf{y}$:

$$\widehat{\boldsymbol{\gamma}}(\omega) = \{\mathbf{X}(\omega)^\mathsf{T}\mathbf{X}(\omega)\}^{-1}\mathbf{X}(\omega)^\mathsf{T}\mathbf{y}.$$

Step 2: the profile sum of squares. Let $\mathbf{P}(\omega) = \mathbf{X}(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\mathbf{X}^\mathsf{T}$ be the projection onto the column space of $\mathbf{X}(\omega)$. Then $\mathbf{X}\widehat{\boldsymbol{\gamma}}(\omega) = \mathbf{P}(\omega)\mathbf{y}$, and since $\mathbf{I} - \mathbf{P}$ is symmetric and idempotent $[(\mathbf{I} - \mathbf{P})^2 = \mathbf{I} - 2\mathbf{P} + \mathbf{P}^2 = \mathbf{I} - \mathbf{P}]$,

$$R(\omega) := \min_{\boldsymbol{\gamma}} Q(\boldsymbol{\gamma}, \omega) = \|\mathbf{y} - \mathbf{P}(\omega)\mathbf{y}\|^2 = \mathbf{y}^\mathsf{T}(\mathbf{I} - \mathbf{P}(\omega))^\mathsf{T}(\mathbf{I} - \mathbf{P}(\omega))\mathbf{y} = \mathbf{y}^\mathsf{T}\mathbf{y} - \mathbf{y}^\mathsf{T}\mathbf{P}(\omega)\mathbf{y}.$$

Step 3: minimize over ω. For every $(\boldsymbol{\gamma}, \omega)$ we have $Q(\boldsymbol{\gamma}, \omega) \geq R(\omega) \geq \min_{\omega'} R(\omega')$, with equality throughout at $\omega = \widehat{\omega}$ and $\boldsymbol{\gamma} = \widehat{\boldsymbol{\gamma}}(\widehat{\omega})$, where

$$\boxed{\widehat{\omega} = \operatorname*{arg\,min}_{\omega \in [0,\pi]} R(\omega) = \operatorname*{arg\,max}_{\omega \in [0,\pi]} \ \mathbf{y}^\mathsf{T}\mathbf{X}(\omega)\{\mathbf{X}(\omega)^\mathsf{T}\mathbf{X}(\omega)\}^{-1}\mathbf{X}(\omega)^\mathsf{T}\mathbf{y}.}$$

Hence $\min_{\boldsymbol{\gamma},\omega} Q = \min_\omega R(\omega)$, and the least-squares estimators are $\widehat{\omega}$ together with $\widehat{\boldsymbol{\gamma}}(\widehat{\omega})$. The five-dimensional problem has become a one-dimensional problem in ω on the compact interval $[0, \pi]$; the four linear parameters have been *profiled out*. (At the finitely many ω where $\mathbf{X}(\omega)$ is rank-deficient, see (c), $\mathbf{P}(\omega)$ is still the projection onto the column space, computed with a generalized inverse, so $R(\omega)$ is defined everywhere.)

Solving the one-dimensional problem in practice. $R(\omega)$ is very wiggly: as ω changes by Δ, the phase ωt^2 changes by up to $n^2\Delta$. A local optimizer started at an arbitrary point will usually stop at a local minimum. The standard approach is to evaluate R on a grid of $[0, \pi]$ with spacing much smaller than π/n^2, take the best grid point, and refine it locally

2.2 (b) The other estimators in terms of $\widehat{\omega}$

Write $\widehat{c}_t = \cos(\widehat{\omega}t^2)$, $\widehat{s}_t = \sin(\widehat{\omega}t^2)$ and $\widehat{\mathbf{X}} = \mathbf{X}(\widehat{\omega})$. From (a),

$$\widehat{\boldsymbol{\gamma}} = \begin{pmatrix} \widehat{\alpha} \\ \widehat{\beta} \\ \widehat{A} \\ \widehat{B} \end{pmatrix} = (\widehat{\mathbf{X}}^\mathsf{T}\widehat{\mathbf{X}})^{-1}\widehat{\mathbf{X}}^\mathsf{T}\mathbf{y},$$

that is, $(\widehat{\alpha}, \widehat{\beta}, \widehat{A}, \widehat{B})$ is the solution of the 4×4 linear system (all sums over $t = 1, \ldots, n$)

$$\begin{pmatrix} n & \sum t & \sum \widehat{c}_t & \sum \widehat{s}_t \\ \sum t & \sum t^2 & \sum t\widehat{c}_t & \sum t\widehat{s}_t \\ \sum \widehat{c}_t & \sum t\widehat{c}_t & \sum \widehat{c}_t^2 & \sum \widehat{c}_t\widehat{s}_t \\ \sum \widehat{s}_t & \sum t\widehat{s}_t & \sum \widehat{c}_t\widehat{s}_t & \sum \widehat{s}_t^2 \end{pmatrix} \begin{pmatrix} \widehat{\alpha} \\ \widehat{\beta} \\ \widehat{A} \\ \widehat{B} \end{pmatrix} = \begin{pmatrix} \sum y_t \\ \sum t y_t \\ \sum \widehat{c}_t y_t \\ \sum \widehat{s}_t y_t \end{pmatrix},$$

with $\sum t = n(n+1)/2$ and $\sum t^2 = n(n+1)(2n+1)/6$. The error variance is estimated by $\widehat{\sigma}^2 = R(\widehat{\omega})/n$ (maximum likelihood under normality) or $R(\widehat{\omega})/(n-5)$ (five parameters were estimated).

2.3 (c) For which ω can the other parameters be estimated?

With ω known the model $\mathbf{y} = \mathbf{X}(\omega)\boldsymbol{\gamma} + \boldsymbol{\varepsilon}$ is an ordinary linear model. The least-squares estimator of $\boldsymbol{\gamma}$ is unique, and $\boldsymbol{\gamma}$ is identifiable, if and only if $\mathbf{X}(\omega)$ has full column rank 4, that is, if and only if

$$a + bt + A'c_t + B's_t = 0 \quad \text{for } t = 1, \ldots, n \qquad \Longrightarrow \qquad a = b = A' = B' = 0. \tag{3}$$

(If $\mathbf{X}(\omega)\mathbf{v} = \mathbf{0}$ for some $\mathbf{v} \neq \mathbf{0}$, then $\boldsymbol{\gamma}$ and $\boldsymbol{\gamma} + \mathbf{v}$ give the same mean vector, so they cannot be distinguished by any amount of data.) We need $n \geq 4$.

Proposition 1. *(i) For $\omega \in \{0,\ \pi/2,\ 2\pi/3,\ \pi\}$, $\mathbf{X}(\omega)$ is rank-deficient for every n: not all of α, β, A, B can be estimated.*
(ii) For every other $\omega \in [0, \pi]$, $\mathbf{X}(\omega)$ has full column rank for all sufficiently large n. If $\omega = 2\pi p/q$ with p, q coprime integers, $n \geq q + 1$ suffices.

[Part (ii) of Proposition 1 is not important for Quiz 2 or end-sem exam. Feel free to skip it]

So, in large samples, **all four parameters can be estimated for every $\omega \in [0, \pi]$ except $\omega = 0,\ \pi/2,\ 2\pi/3,\ \pi$.**

Proof of (i): the four exceptional frequencies. $\omega = 0$**.** $c_t = \cos 0 = 1$ and $s_t = \sin 0 = 0$ for all t. The cosine column equals the intercept column and the sine column is zero, so $\operatorname{rank} \mathbf{X} = 2$. The mean is $(\alpha + A) + \beta t$: only $\alpha + A$ and β are estimable, and B does not appear at all.

$\omega = \pi$**.** t^2 has the same parity as t, so $c_t = \cos(\pi t^2) = (-1)^t$ and $s_t = \sin(\pi t^2) = 0$. The sine column is zero and $\operatorname{rank} \mathbf{X} = 3$: α, β, A are estimable, B is not.

$\omega = \pi/2$**.** If t is even, $t^2 = 4m$ and $\omega t^2 = 2\pi m$, so $(c_t, s_t) = (1, 0)$. If t is odd, $t^2 = 4m+1$ and $\omega t^2 = 2\pi m + \pi/2$, so $(c_t, s_t) = (0, 1)$. Hence $c_t + s_t = 1$ for every t: the columns satisfy $\mathbf{c} + \mathbf{s} = \mathbf{1}$, and $\operatorname{rank} \mathbf{X} = 3$. The mean can be written $(\alpha + A)c_t + (\alpha + B)s_t + \beta t$, so only $\alpha + A$, $\alpha + B$ and β are estimable.

$\omega = 2\pi/3$**.** The squares modulo 3 are 0 (when $3 \mid t$) and 1 (otherwise). So $\omega t^2 \equiv 0$ or $2\pi/3 \pmod{2\pi}$, and (c_t, s_t) equals $(1, 0)$ or $(-\frac{1}{2}, \frac{\sqrt{3}}{2})$. Both points satisfy $c + \sqrt{3}\, s = 1$, so $\mathbf{c} + \sqrt{3}\,\mathbf{s} = \mathbf{1}$, and $\operatorname{rank} \mathbf{X} = 3$.

In each case the points (c_t, s_t), $t = 1, \ldots, n$, take at most two distinct values on the unit circle, and any two points lie on a line $a + A'x + B'y = 0$; this gives a non-trivial solution of (3) with $b = 0$. □

Proof of (ii). The geometric fact behind the proof is that a straight line meets the unit circle in at most two points. If a relation (3) holds with $b = 0$ and $(A', B') \neq (0, 0)$, then every point $P_t = (c_t, s_t)$ lies on the line $a + A'x + B'y = 0$; so if the P_t take at least three distinct values, we must have $A' = B' = 0$, and then $a = 0$.

Rational case: $\omega = 2\pi p/q$, $\gcd(p,q) = 1$. Since $0 < \omega \le \pi$ we have $1 \le p$ and $2p \le q$. The exceptional cases are $q = 2$ ($\omega = \pi$), $q = 3$ ($\omega = 2\pi/3$) and $q = 4$ ($\omega = \pi/2$), so assume $q \ge 5$ and $n \ge q + 1$.

- *Periodicity forces* $b = 0$. $\omega(t+q)^2 - \omega t^2 = \omega q(2t+q) = 2\pi p(2t+q) \in 2\pi\mathbb{Z}$, so $c_{t+q} = c_t$ and $s_{t+q} = s_t$. Subtracting (3) at t from (3) at $t+q$ (both $\le n$ for $t = 1$) gives $bq = 0$, so $b = 0$.
- *Three distinct points.* Take $t = q$, 1, 2 (all $\le n$). The angles are $\omega q^2 = 2\pi pq \equiv 0$, $\omega = 2\pi p/q$ and $4\omega = 2\pi \cdot 4p/q \pmod{2\pi}$. They are distinct if $q \nmid p$, $q \nmid 4p$ and $q \nmid 3p$; since $\gcd(p,q) = 1$ this means $q \nmid 1$, $q \nmid 4$, $q \nmid 3$, which holds because $q \ge 5$. So P_q, P_1, P_2 are three distinct points on the circle, and by the geometric fact $A' = B' = a = 0$.

Irrational case: ω/π *irrational.* First, the points P_t, $t = 1, 2, \dots$, are all distinct: $P_t = P_s$ with $t \ne s$ would mean $\omega(t^2 - s^2) \in 2\pi\mathbb{Z}$, making ω/π rational. Next, if $\mathbf{X}(\omega)$ is rank-deficient for some n, it is rank-deficient for every smaller n (a relation for $t \le n$ is also a relation for $t \le m$). So if the claim failed, $\mathbf{X}(\omega)$ would be rank-deficient for *every* n. Choose for each n a unit vector $\mathbf{v}_n = (a_n, b_n, A'_n, B'_n)$ satisfying (3) for $t \le n$. By compactness of the unit sphere a subsequence converges to a unit vector $\mathbf{v} = (a, b, A', B')$, and since each equation of (3) is continuous in $\mathbf{v}$ and is satisfied by $\mathbf{v}_n$ once $n \ge t$, the limit satisfies $a + bt + A'c_t + B's_t = 0$ for *all* $t \ge 1$. The term $A'c_t + B's_t$ is bounded by $\sqrt{A'^2 + B'^2}$, so $a + bt$ is bounded in t, forcing $b = 0$. Then infinitely many distinct points P_t lie on the line $a + A'x + B'y = 0$, forcing $A' = B' = 0$ and $a = 0$, which contradicts $\|\mathbf{v}\| = 1$. □

Remark 2. Near the exceptional frequencies $\mathbf{X}(\omega)$ is nearly singular, and the estimators of the affected parameters have very large variances. With $n = 50$, the smallest singular value of $\mathbf{X}(\omega)$ (columns scaled to unit length) is 0 at $\omega = 0$, 8.8×10^{-4} at $\omega = 10^{-4}$ and 0.33 at $\omega = 0.01$; at a typical frequency such as $\omega = 1$ it is about 0.36.

2.4 (d) A 95% confidence set for (α, β, A, B), ω known

Assume ω is known and is not one of the exceptional values, so $\mathbf{X} = \mathbf{X}(\omega)$ has full rank 4. For exact results assume also that the errors are normal, $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2\mathbf{I})$.

Step 1: distribution of $\widehat{\boldsymbol{\gamma}}$. Since $\widehat{\boldsymbol{\gamma}} = (\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\mathbf{X}^\mathsf{T}\mathbf{y} = \boldsymbol{\gamma} + (\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\mathbf{X}^\mathsf{T}\boldsymbol{\varepsilon}$,

$$\mathbb{E}\widehat{\boldsymbol{\gamma}} = \boldsymbol{\gamma}, \qquad \operatorname{Cov}(\widehat{\boldsymbol{\gamma}}) = (\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\mathbf{X}^\mathsf{T}(\sigma^2\mathbf{I})\mathbf{X}(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1} = \sigma^2(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1},$$

$$\widehat{\boldsymbol{\gamma}} \sim N_4\big(\boldsymbol{\gamma}, \sigma^2(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\big).$$

Step 2: two independent chi-squares. With $\mathbf{P} = \mathbf{X}(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\mathbf{X}^\mathsf{T}$,

$$(\widehat{\boldsymbol{\gamma}} - \boldsymbol{\gamma})^\mathsf{T}\mathbf{X}^\mathsf{T}\mathbf{X}(\widehat{\boldsymbol{\gamma}} - \boldsymbol{\gamma}) = \boldsymbol{\varepsilon}^\mathsf{T}\mathbf{X}(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\mathbf{X}^\mathsf{T}\mathbf{X}(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\mathbf{X}^\mathsf{T}\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}^\mathsf{T}\mathbf{P}\boldsymbol{\varepsilon},$$

$$\mathrm{RSS} = \|\mathbf{y} - \mathbf{X}\widehat{\boldsymbol{\gamma}}\|^2 = \boldsymbol{\varepsilon}^\mathsf{T}(\mathbf{I} - \mathbf{P})\boldsymbol{\varepsilon}.$$

$\mathbf{P}$ and $\mathbf{I} - \mathbf{P}$ are symmetric idempotent matrices of ranks 4 and $n - 4$, and $\mathbf{P}(\mathbf{I} - \mathbf{P}) = \mathbf{0}$. Hence $\boldsymbol{\varepsilon}^\mathsf{T}\mathbf{P}\boldsymbol{\varepsilon}/\sigma^2 \sim \chi^2_4$ and $\boldsymbol{\varepsilon}^\mathsf{T}(\mathbf{I} - \mathbf{P})\boldsymbol{\varepsilon}/\sigma^2 \sim \chi^2_{n-4}$, and they are independent (the normal vectors $\mathbf{P}\boldsymbol{\varepsilon}$ and $(\mathbf{I} - \mathbf{P})\boldsymbol{\varepsilon}$ have zero covariance $\sigma^2\mathbf{P}(\mathbf{I} - \mathbf{P})$).

Step 3: the pivot. With $s^2 = \mathrm{RSS}/(n-4)$,

$$F(\boldsymbol{\gamma}) = \frac{(\widehat{\boldsymbol{\gamma}} - \boldsymbol{\gamma})^{\mathsf{T}} \mathbf{X}^{\mathsf{T}} \mathbf{X} (\widehat{\boldsymbol{\gamma}} - \boldsymbol{\gamma})/4}{s^2} = \frac{\{\boldsymbol{\varepsilon}^{\mathsf{T}} \mathbf{P} \boldsymbol{\varepsilon}/\sigma^2\}/4}{\{\boldsymbol{\varepsilon}^{\mathsf{T}} (\mathbf{I} - \mathbf{P}) \boldsymbol{\varepsilon}/\sigma^2\}/(n-4)} \sim F_{4,\,n-4},$$

whose distribution does not depend on any unknown parameter. Therefore $P\{F(\boldsymbol{\gamma}) \le F_{4,n-4,0.95}\} = 0.95$, and the 95% confidence set is the ellipsoid

$$\boxed{C = \Big\{(\alpha, \beta, A, B) \in \mathbb{R}^4 : \ (\widehat{\boldsymbol{\gamma}} - \boldsymbol{\gamma})^{\mathsf{T}} \mathbf{X}(\omega)^{\mathsf{T}} \mathbf{X}(\omega) (\widehat{\boldsymbol{\gamma}} - \boldsymbol{\gamma}) \le 4\, s^2\, F_{4,\,n-4,\,0.95}\Big\},}$$

centred at $\widehat{\boldsymbol{\gamma}}$, with exact coverage 95% under normal errors.